

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459404.9471 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.011
Transit depth (Rp/Rs)^2	3.24 +/- 0.39 [%]
Orbital Inclination (inc)	87.75 +/- 0.78
Airmass coefficient 1 (a1)	0.972 +/- 0.011
Airmass coefficient 2 (a2)	0.024 +/- 0.022
Scatter in the residuals of the lightcurve fit is	1.14 %
Best Comparison Star	#3 - [143, 429]
Optimal Aperture	5.15
Optimal Annulus	8.59
Transit Duration (day)	0.0733 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.18 $\pm$ 0.011 vs 0.1714 (deviation 0.64 σ)
Duration fit vs published	0.0733 d vs 0.0892 d (deviation 3.02 σ)
Mid-time O-C	-7.6 min
Depth SNR	13.4
Residual scatter	1.14 %

Light curve

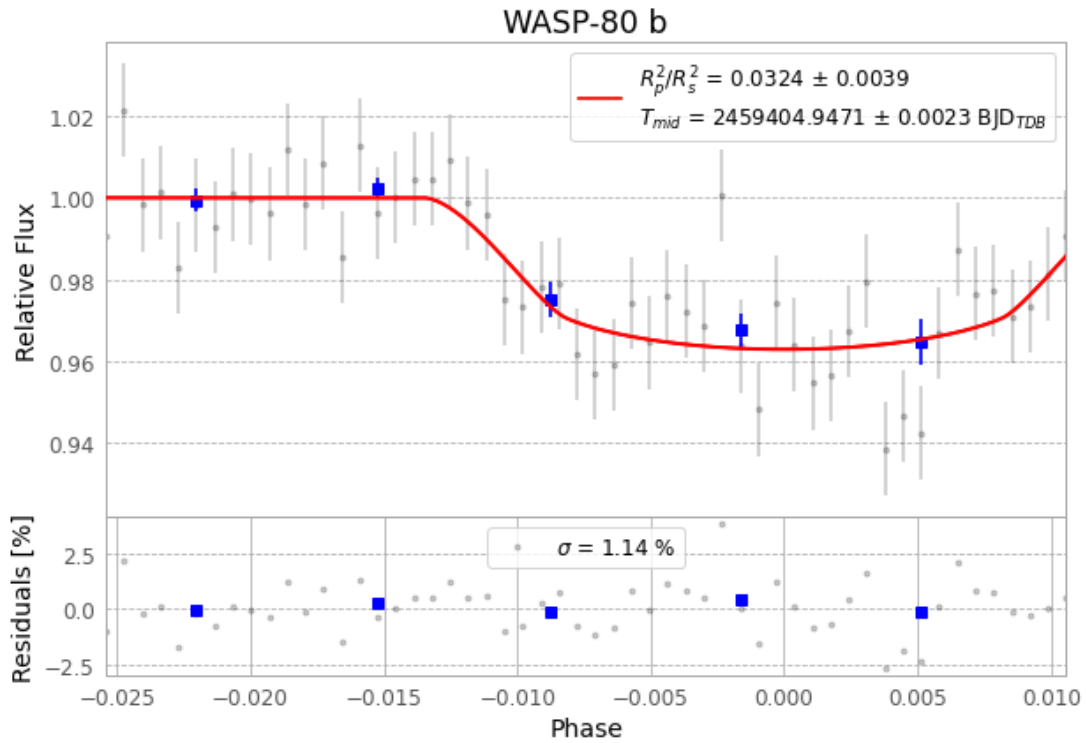


Figure 1 — FinalLightCurve_WASP-80 b_2021-07-09.png (SHA-256 5cc52a3e01b3ef5d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [386, 102], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6c8e06f809b31530...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

55 FITS frames (36 MB), WASP-80210709084212.FITS → WASP-80210709112412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-80-210709.log (bfa114db789c...), FinalLightCurve_WASP-80 b_2021-07-09.png (5cc52a3e01b3...), FinalLightCurve_WASP-80 b_2021-07-09.csv (83d5a0b44932...)

Compute resources

Wall time 128.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 23:41Z.